Does Trust Decohere? A Pre-Registered Test for Quantum Hardware Reliability

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***Abstract*—A qubit loses its state through decoherence, and a claim about that qubit, such as yesterday's calibration, loses its validity in a formally similar way. We ask whether that analogy is operationally useful. Lindblad Trust Networks (LTN) are a temporal graph model whose per-node trust states are density matrices evolving under learned Lindblad master equations, with rates coupled to measured device physics, updated and propagated by completely positive maps, and read out by learned positive-operator-valued measurements. We evaluate pre-registered hypotheses on public telemetry from three real platforms and a labeled synthetic fleet. The verdict is negative and, we argue, instructive. LTN does not beat capacity-tuned recurrent, temporal-graph, and transformer forecasters: on real telemetry from three IBM Eagle devices over 14 days it ranks third of the four learned models, ahead only of the transformer, and no ablation, including removal of the density matrix itself, reproducibly changes performance. Divergence anticipation, zero-shot cross-platform ranking, and uncertainty-aware selection all fail their pre-registered criteria. Pretraining does yield a large cold-start gain, but under an identical protocol every sequence baseline transfers better, so the gain belongs to pretraining rather than to the representation. A state-sufficiency test explains why: zeroing the recurrent carry at readout costs over 80 % NCRPS while permuting the density matrix costs 4 % on synthetic and 0.07 % on real telemetry, so forecasts route around the physics-structured state. Uncertainty is calibrated in simulation but under-dispersed on real data, where the trained state also grows *more* confident under evidence starvation, the opposite of the design intent. We report what a physics-structured trust state would have to do differently to earn its place. We release the schema, the frozen benchmark, and one-command reproduction from public data.**

***Index Terms*—device calibration, Lindblad dynamics, quantum computing, temporal graph learning, trust evaluation, uncertainty quantification**

Nomenclature

Acronyms

CPTP	Completely positive, trace preserving.
CRPS	Continuous ranked probability score.
DAQEC	Public IBM device-telemetry archive.
ECE	Expected calibration error.
LCB	Lower confidence bound.
LER	Logical error rate.
LTN	Lindblad trust network.
NCRPS	Normalized CRPS (primary forecasting metric).
POVM	Positive operator-valued measure.
QEC	Quantum error correction.
TGN	Temporal graph network.
UTS	Unified telemetry schema.

Trust state and its dynamics

ρ	Trust state; $\rho \succeq 0$, $\operatorname{tr}\rho = 1$.
d	State dimension, $\rho \in \mathbb{C}^{d\times d}$.
$\mathcal{L}$	Lindblad generator (superoperator).
H	Hamiltonian (Hermitian) part of $\mathcal{L}$.
$L_k,\ \gamma_k$	Jump operators and their learned rates.
Δt	Elapsed time since the last evidence event.
τ	Saturating clock, $\tau_{\max}(1 - e^{-\Delta t/\tau_{\max}})$.

Evidence and readout

K_i	Kraus operators, $\sum_i K_i^\dagger K_i = I$.
E_c	POVM element, $\sum_c E_c = I$.
$S(\rho)$	Von Neumann entropy.
$v(\rho)$	Vacuity, $\frac{d}{d-1}(1 - \operatorname{tr}\rho^2)$; absence of evidence.
μ, σ	Predictive mean and scale of the Gaussian head.

Forecasting and evaluation

h	Forecast horizon.
Q_{t+h}	Device-quality observable at horizon h.
q	Specification limit; violation is Q_{t+h} out of spec.
$R(h)$	Reliability function, P(no violation through h).
δ	Cliff's delta (effect size).
k	Days of target-device data in cold-start transfer.

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I. Introduction

Quantum hardware is a computing substrate whose failure physics and whose trustworthiness dynamics admit a common mathematical description. A qubit loses its state to the environment by decoherence, at rates $1/T_1$, $1/T_2$ measured every calibration cycle. A *claim about* that qubit, such as yesterday's published T_1, gate error, or readout fidelity, loses its validity the same way, gradually and stochastically, as the device drifts away from the snapshot. Today the first process is exquisitely characterized while the second is not modeled at all: cloud users pick backends and qubit subsets from calibration data that is hours to days old, QKD operators assign the label *trusted* to relay nodes as a static, binary, human choice, and no continuous, evidence-driven, uncertainty-aware trust layer exists for quantum infrastructure. This gap persists even though trust evaluation is a mature ML subfield on social and transaction graphs, and even though quantum telemetry is uniquely rich and public. The cost is concrete: jobs placed on drifting qubit subgraphs waste shots; silent divergence between claimed and delivered device quality is invisible until results fail.

We introduce *Lindblad Trust Networks* (LTN): a temporal graph model over quantum devices whose per-node trust states are density matrices evolving under *learned Lindblad master equations*, with epistemic decoherence rates coupled to the devices' measured physical decoherence (T_1, T_2, calibration age). Between observations trust decays toward the maximally mixed (vacuous) state at learned, physics-informed rates; evidence enters through learned CPTP channels; propagation over the coupling graph is completely-positive message passing; and readout is by learned POVMs, with two complementary uncertainty indicators, *vacuity* (absence of evidence) and *dissonance* (conflict of evidence). The hypothesis is compact:

trust should decohere without evidence, at rates we can learn. We test it and, on real telemetry, it does not hold: the fitted state sharpens rather than mixes as it extrapolates (§V-G), and the physics-coupled rates are not reproducibly load-bearing (Table IV).

We evaluate three hypotheses on public real-hardware telemetry from three platforms (IBM superconducting devices via the DAQEC archive, Google quantum-error-correction releases, silicon spin qubits), complemented by an explicitly labeled synthetic fleet calibrated to the real marginals. **H1 (dynamics)** holds that the Lindblad inductive bias beats budget-matched neural baselines at probabilistic telemetry forecasting. **H2 (transfer)** holds that pretraining yields few-shot adaptation to unseen devices and nontrivial zero-shot reliability ranking on a held-out platform. **H3 (trust)** holds that the trust state is calibrated, that its dissonance anticipates claim-versus-delivery divergence, and that its uncertainty converts to lower selection regret than calibration-greedy policies. The verdict is negative and, we argue, instructive. LTN does *not* beat budget-matched sequence models at forecasting, because on three IBM devices over 14 days it ranks third of four learned models, and no module ablation reproducibly improves it. Pre-registered cold-start transfer shows a large few-shot gain, but a post-hoc control in which every sequence baseline receives the identical pretraining protocol shows that the gain belongs to pretraining, not to the density-matrix representation (§V-D). Predictive uncertainty is calibrated in simulation but under-dispersed on real data, and the trust-semantic hypotheses (zero-shot transfer, divergence anticipation, entropy-growth signature) yield clean, mechanistically explained negatives. Every negative is reported against its pre-registered criterion, including one where a single diverging seed manufactures every ablation's apparent advantage, which we report as such rather than claim as a result. We then treat our own mechanisms as hypotheses and test them, by building the model they imply (§V-G) and by measuring the quantities they predict. Three of the four outcomes are negative. The variance is causally attributed to the learned Hamiltonian by a paired ablation, but the non-normality mechanism we proposed for it is contradicted by the generator spectra and we withdraw it (§V-A); the readout diagnosis survives only in part; the representational one fails; and a direct state-sufficiency test (§V-I) shows the forecast reaches the state mainly through the recurrent carry rather than through the density matrix.

Contributions. (i) *Quantum subjective logic*: building on the Bayesian reading of density matrices [5], we couple density-matrix trust opinions to physics-informed Lindblad dynamics and give classical-recovery, fusion non-commutativity, and no-amplification properties (§III); (ii) *architecture*: six named modules (DM-State, physics-coupled Neural Lindblad drift, CPTP evidence channels, KrausProp message passing, POVM readout, and a spectral vacuity and dissonance readout), four of which we ablate individually in E2; to our knowledge this combination has not appeared in published trust models or quantum GNNs; (iii) *system and artifacts*: a universal telemetry schema spanning five sources, a token-ready IBM fleet harvester, frozen benchmark splits, and one-command reproduction from public data; (iv) *application*: trust-aware qubit-subgraph selection with a risk-averse lower-confidence-bound policy, evaluated by replay regret; (v) *falsified diagnoses*: rather than close with mechanisms we merely assert, we implement the model our two architectural diagnoses imply and re-run the affected experiments under the identical protocol, confirming one and refuting the other (§V-G). The confirmatory core of this study is pre-registered: frozen splits, success bars and tuning budgets were committed before any training run, under the tags recorded in the deviation register. The paper additionally reports a set of *post-hoc* analyses, added after the pre-registered results were known, and each is labelled as such where it appears (§V-B, §V-C, §V-D, §V-H, §V-I, §V-G). Readers should weight the two differently: the pre-registered arms test hypotheses fixed in advance, whereas the post-hoc analyses were designed once their targets were visible and are best read as diagnostics rather than as confirmations. All reported numbers are produced by the accompanying evaluation pipeline from logged runs.

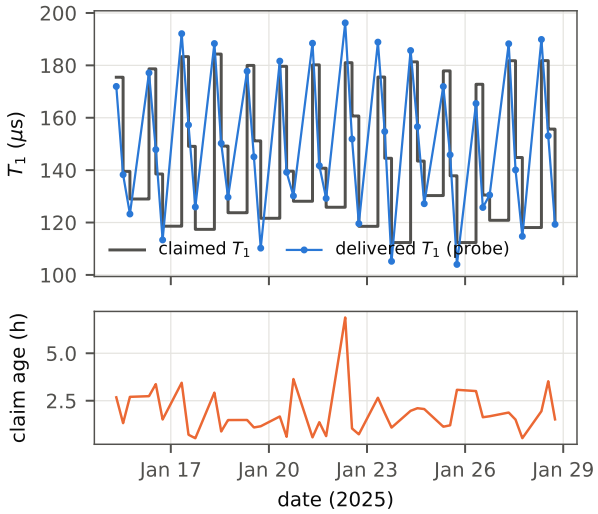


Fig. 1. The two decoherences on one real device (`ibm_brisbane`, 14 days). Top: probe-measured T_1 drifts while the claimed calibration value goes stale between refreshes. Bottom: the age of the claim, the evidence whose decay LTN models. Claim-versus-probe discrepancy correlates with claim age ($r = 0.34$, $n = 42$, 95% CI $[0.04, 0.58]$).

II. Related Work

Trust evaluation on graphs. Graph neural trust models such as TrustGNN [2], EvolveTrust [3], DLGTrust [4], and the dynamic, uncertainty-aware DGTEN [1] operate on social or Bitcoin transaction graphs with real-vector or Gaussian node states and generic learned dynamics (DGTEN, closest to our setting, uses Gaussian node states with ODE-based residual updates). None addresses physical infrastructure, none has a physics-informed evolution law, and none represents evidence through quantum channels. LTN shares DGTEN's dynamic, uncertainty-aware framing but moves to a new state space (density matrices), a new dynamics law (learned Lindbladians), and a new domain (quantum hardware telemetry). E8 evaluates the DM-state on Bitcoin-OTC and Bitcoin-Alpha [11] against an equal-parameter vector control to show the representation is not domain-bound; it is an internal architecture ablation, not a re-benchmarking of DGTEN.

Quantum graph neural networks. The QGNN literature quantizes the *model*, using parameterized quantum circuits as

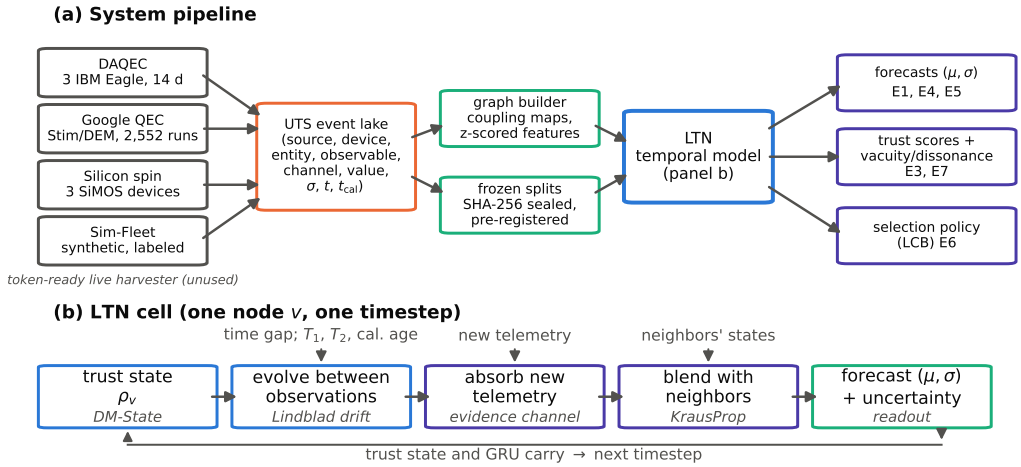


Fig. 2. System overview. (a) End-to-end pipeline: four public telemetry sources (plus a token-ready live harvester, unused here) are normalized into the universal telemetry event lake, converted by the graph builder into per-device temporal graphs under SHA-256-sealed pre-registered splits, and consumed by the LTN temporal model, whose outputs feed the three experiment families (forecasting, trust semantics, selection). (b) One LTN cell for node v at one timestep: the trust state evolves over the time since the last observation, at rates set by the device's T_1, T_2 and calibration age; absorbs the new telemetry; blends with its graph neighbors' states; and is read out as trust scores, a Gaussian forecast, and the entropy, vacuity and dissonance summaries. The state and a GRU carry pass to the next timestep. Italics name each stage's module (§III), whose validity guarantee is enforced by the accompanying unit tests.

layers, to process classical data. LTN occupies the inverse, previously empty quadrant: a classical model with a quantum-structured *representation*, applied to quantum-*generated* data. No quantum computation is claimed or required (§VI).

Drift characterization and digital twins. Calibration-drift studies and digital-twin models [19], [21] forecast single-device parameters with generic ML; they carry no graph structure, no trust semantics, no claim-versus-delivery distinction, and no cross-platform transfer. LTN subsumes their forecasting task (E1) and adds the trust layer they lack (E3, E6, E7).

Trusted nodes in QKD and verification. QKD network security models treat relay trust as a static Boolean or consume labels without learning [20]; cryptographic verification of quantum clouds is per-job and complementary, because a continuous trust layer can budget expensive verification toward low-trust targets, which we leave to future work.

To our knowledge, no prior work combines physics-coupled learned open-system dynamics over a temporal device graph, training on multi-platform real telemetry, and calibrated claim-versus-delivery trust semantics.

III. The LTN Framework

LTN maintains, for every node v of a temporal device graph (qubits, couplers, qubit subsets, or whole devices), a *trust state*: a density matrix $\rho_v \in \mathbb{C}^{d\times d}$, $\rho_v \succeq 0$, $\operatorname{tr}\rho_v = 1$ (default d=4). The maximally mixed state $\rho = I/d$ is the vacuous prior of total ignorance, and every operation below maps valid states to valid states by construction, a property we unit-test rather than assume.

A. Universal telemetry schema

All five data sources (Table II) are mapped to a single event schema: $(\text{source}, \text{platform}, \text{device}, \text{entity}, \text{observable}, \text{channel}, \text{value}, \sigma, t, t_{\text{cal}})$, where *channel* distinguishes claimed calibration values from probe-measured and delivered outcomes, giving the claim-versus-delivery pairing that trust evaluation requires.

B. DM-State

Node states are parameterized by unconstrained $\theta_v \in \mathbb{R}^{d^2}$ via a Cholesky factor: $\rho_v = L_v L_v^\dagger / \operatorname{tr}(L_v L_v^\dagger)$ with L_v lower triangular (positive real diagonal). PSD and unit trace hold for any θ_v, and gradients flow through the factorization.

C. Lindblad drift: epistemic decoherence

Between observations, trust states evolve under a *learned* Lindblad master equation [7], the same dynamical form that governs the devices' physical decoherence:

$$\dot\rho \;=\; -i[H_v,\rho] \;+\; \sum_{k=1}^{3}\gamma_{v,k}\Big(L_k\rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k,\rho\}\Big)\,, \quad (1)$$

with H_v Hermitian from the node embedding and $\{L_k\}$ a shared learned jump dictionary (initialized to lowering, dephasing, and a cyclic-shift mixing operator). The physics coupling at the heart of H1 sets the rates,

$$\gamma_{v,k} \;=\; \operatorname{softplus} g_\theta\Big(T_1^{(v)}, T_2^{(v)}, \Delta t_{\text{cal}}, \ell_v, e_{\text{plat}}\Big)\,, \quad (2)$$

so epistemic decoherence rates are functions of the node's *measured physical* coherence times and the age of its calibration evidence, with $\partial\gamma_k/\partial\Delta t_{\text{cal}} > 0$ enforced at initialization: stale claims decay. Over a gap Δt we integrate (1) in closed form, vectorizing the generator to a $d^2\times d^2$ superoperator (column stacking, so that $\operatorname{vec}(A\rho B) = (B^\top\otimes A)\operatorname{vec}(\rho)$) and applying a batched matrix exponential. The resulting channel is CPTP for any parameters (trace error $< 10^{-8}$, Choi minimum eigenvalue $> -10^{-7}$), and for general Hermitian H at d=4 it reproduces an independent `qutip` Lindblad integration to $< 10^{-9}$; both are checked in the accompanying tests. At initialization the unital (mixing) dissipators dominate, so evidence-starved states drift toward the maximally mixed vacuous state: trust decays toward *ignorance*, never toward unwarranted confidence.

D. Evidence channels and KrausProp

A telemetry event with features $x \in \mathbb{R}^m$ (per-observable z-scores, missingness mask, calibration age, platform) updates the state through a learned CPTP channel: an MLP

emits $r{=}4$ matrices $\{A_i\}$, normalized to Kraus operators $K_i = A_i M^{-1/2}$ with $M = \sum_i A_i^\dagger A_i$, giving $\rho \leftarrow \sum_i K_i \rho K_i^\dagger$ with $\sum_i K_i^\dagger K_i = I$ exactly. Messages between graph neighbors use the same channel construction conditioned on edge features (two-qubit gate errors); aggregation is a convex mixture $\rho_v \leftarrow (1-\beta_v)\rho_v + \beta_v \sum_u \alpha_{uv}\, m_{u\to v}$ with attention $\alpha_{uv} \propto \exp(\mathrm{Re}\ \mathrm{tr}(\rho_u \rho_v)/\tau)$. Convexity keeps states valid, and the non-commutativity of channel composition (Proposition 2) lives in multi-hop propagation.

E. Readout and uncertainty summaries

Trust facets are learned POVMs: $E_c = C_c^\dagger C_c$ set-normalized so $\sum_c E_c = I$; $\mathrm{score}_c = \mathrm{tr}(E_c\rho) \in [0,1]$. Forecasts are Gaussian (μ, σ) read from the state *after an evidence-free Lindblad roll to the requested horizon*, so predictive uncertainty inherits the physics of evidence decay (Fig. 8). Overall uncertainty is the von Neumann entropy $S(\rho)$, the pre-registered E7 signal; we additionally report two qualitatively distinct, individually normalized summaries of *why* the state is uncertain: *vacuity* $v(\rho) = \frac{d}{d-1}(1 - \mathrm{tr}\,\rho^2) \in [0,1]$, the purity deficit that captures *absence* of evidence ($v{=}1$ at the maximally mixed state), and *dissonance*, the off-diagonal coherence mass $\big(\sum_{i\neq j} |\rho_{ij}|\big)/(d{-}1) \in [0,1]$ that captures *conflicting* evidence. These are the density-matrix analogues of the uncertainty and conflict masses in subjective logic [6]; unlike a scalar entropy they separate the two sources, but they are complementary indicators, not an additive decomposition of $S(\rho)$: vacuity, like $S(\rho)$, measures mixedness; dissonance is basis-dependent and can vary at fixed $S(\rho)$.

F. Quantum subjective logic

Interpreting a density matrix as an agent's degree of belief is the premise of quantum Bayesianism [5]; our contribution is not that reading but its coupling to physics-informed Lindblad trust dynamics. The properties below anchor the semantics. They are stated under the following assumptions.

1) Evidence incorporation and graph propagation act on the trust state by completely positive trace-preserving (CPTP) maps.
2) The *discounting* channel of Proposition 3 is unital, $\Phi(I/d) = I/d$. The learned dissipators are in general non-unital (e.g. the lowering-type L_0), so Assumption 2 restricts that proposition to the discounting sub-class rather than to the full trained model.
3) Within an evidence gap the Lindblad generator is time-local and held constant, so propagation over the gap is a one-parameter semigroup.
4) The forecast head is Gaussian: predictive uncertainty is summarized by (μ, σ) rather than by the full predictive law.
5) Telemetry absent within an injected gap is missing at random with respect to the observable being forecast.

Proposition 1 (Classical embedding)**.** *Let* $\rho = \mathrm{diag}(p_1, \ldots, p_d)$. *Then (i) the POVM scores reduce to affine functionals* $\mathrm{score}_c = \sum_j (E_c)_{jj}\, p_j$ *of* p *and, by set-normalization, form a probability vector; (ii)* $(\mathrm{score}_c)_c$ *and the vacuity* $v(\rho)$ *are* not *jointly normalized, but the rescaling* $b_c := (1-v)\,\mathrm{score}_c$, $u := v$, *with base rate* $a_c := \mathrm{score}_c$, *is a bona fide subjective-logic opinion [6] whose projected probability* $b_c + a_c u$ *returns* score_c*; (iii) the correspondence is an* embedding, *not a bijection, because a diagonal* d*-dimensional state carries* $d-1$ *free parameters against a multinomial opinion's* $2d-1$, *so the model realizes a lower-dimensional family inside the opinion simplex; and (iv) on diagonal states the evidence channel acts as a column-stochastic (Markov) transition on* p.

Proposition 2 (Fusion non-commutativity)**.** *Fusion of two evidence channels is order-independent* if *they admit Kraus representations* $\{K_i\}, \{L_j\}$ *with* $[K_i, L_j]\, P_\rho = 0$ *for all* i, j, *where* P_ρ *projects onto* $\mathrm{supp}(\rho)$. *The condition is sufficient but not necessary.*[1]

Proposition 3 (No amplification, unital case)**.** *Under Assumption 2, a unital discounting channel never* increases *a state's purity:* $\mathrm{tr}\,\Phi(\rho)^2 \le \mathrm{tr}\,\rho^2$, *so* $v(\Phi(\rho)) \ge v(\rho)$ *and the discounted opinion's certainty* ($1-$*vacuity) cannot exceed the source's.*

The proofs are given in the Appendix.

Training minimizes $\mathcal{L} = \lambda_1 \mathrm{NLL}_{\mathrm{Gauss}} + \lambda_2 \mathrm{BCE}_{\mathrm{diverge}} + \lambda_3 \mathrm{recon}_{\mathrm{masked}} + \lambda_4 \mathrm{ECE}_{\mathrm{soft}}$ (Adam, 10^{-3} cosine, gradient clip 1.0, early stopping on validation CRPS); cross-platform pretraining uses λ_3 only (masked-telemetry reconstruction). $\mathrm{ECE}_{\mathrm{soft}}$ is a differentiable expected-calibration error over $B{=}15$ bins: writing w_{ib} for the membership of sample i in bin b, formed from sigmoid kernels $\sigma(\kappa(p_i - e_b))$ at the internal bin edges e_b with sharpness $\kappa{=}50$ and padded so that $\sum_b w_{ib} = 1$ exactly,

$$\mathrm{ECE}_{\mathrm{soft}} = \sum_{b=1}^{B} \frac{m_b}{n} \left|\bar{p}_b - \bar{y}_b\right|, \tag{3}$$

with $m_b = \sum_i w_{ib}$, $\bar{p}_b = \sum_i w_{ib} p_i / m_b$ and $\bar{y}_b = \sum_i w_{ib} y_i / m_b$. The hard ECE is recovered as $\kappa \to \infty$; the soft form keeps the term connected to the autograd graph.

IV. Experimental Setup

Data (Table II). All sources are public. *DAQEC* [21]: 756 QEC runs on three IBM Eagle devices over 14 days at 8-hour cadence, pairing claimed calibration (T_1, T_2, readout, ECR error; calibration timestamps) with probe-measured values and delivered logical error rates per qubit subgraph. We use its telemetry columns only, not its authors' experimental conclusions (§VI). *Google QEC releases* [22], [23], [24], [25]: Stim circuits, detector error models (DEMs), and raw shot data for surface-, color-, and dynamic-code experiments on Sycamore and Willow; we parse the circuits and shots with Stim [17], extract per-qubit DEM error intensities (calibration analog), and decode logical error per round with minimum-weight perfect matching [18] (2,552 runs). *Silicon spin qubits* [26]: CZ-fidelity and T_2 time series for three SiMOS devices, including lab-timestamped fidelity traces. *Sim-Fleet* (synthetic, explicitly labeled): a 16-device, 180-day generative fleet with

[1] Kraus representations are non-unique, and channels may commute as superoperators with no commuting Kraus pair: the fully depolarizing channel commutes with every unitary conjugation, yet its Kraus set spans $\mathbb{M}_d$ in every representation. The representation-independent criterion is commutation of the two Liouville matrices.

TABLE I
SIM-FLEET VALIDATION AGAINST DAQEC: EVERY STATISTIC IN THE FIT OBJECTIVE, REPORTED WHETHER OR NOT IT MATCHED. REL. ERR. IS (SIM − REAL)/REAL. ‡SIGN DIFFERS BETWEEN REAL AND SIMULATED SERIES.

Statistic	Real (DAQEC)	Sim (simfleet-v1)	Rel. err.	Obj.
disc.mean	0.1037	0.1511	+45.7%	✓
disc.p95	0.2836	0.3843	+35.5%	✓
gate_error_2q.mean	0.008017	0.005831	-27.3%	✓
gate_error_2q.q10	0.00657	0.004277	-34.9%	✓
gate_error_2q.q50	0.007967	0.00569	-28.6%	✓
gate_error_2q.q90	0.009627	0.007548	-21.6%	✓
gate_error_2q.std	0.001152	0.001349	+17.1%	✓
readout_error.mean	0.01503	0.01609	+7.0%	✓
readout_error.q10	0.0115	0.008384	-27.1%	✓
readout_error.q50	0.01494	0.0149	-0.3%	✓
readout_error.q90	0.01885	0.02529	+34.2%	✓
readout_error.std	0.002867	0.006999	+144.1%	✓
recal.mean_h	1.313	8.324	+534.0%	✓
t1.acf1‡	-0.156	0.2019	+229.5%	✓
t1.acf3	0.4205	0.1659	-60.5%	✓
t1.persist_mae	0.06406	0.119	+85.7%	✓
t1_us.mean	160.5	180.7	+12.6%	✓
t1_us.q10	120.4	90.66	-24.7%	✓
t1_us.q50	158	165.5	+4.8%	✓
t1_us.q90	204.8	288.5	+40.9%	✓
t1_us.std	31.86	84.88	+166.4%	✓
t2_us.mean	89.7	106.8	+19.1%	✓
t2_us.q10	68.7	65.44	-4.8%	✓
t2_us.q50	88.49	101.5	+14.7%	✓
t2_us.q90	112.4	154.8	+37.7%	✓
t2_us.std	16.87	36.5	+116.4%	✓
KS t1_us	$D = 0.2235$, $p < 10^{-16}$		–	–
KS t2_us	$D = 0.2871$, $p < 10^{-16}$		–	–
KS readout_error	$D = 0.2078$, $p < 10^{-16}$		–	–
KS gate_error_2q	$D = 0.6546$, $p < 10^{-16}$		–	–

Ornstein–Uhlenbeck coherence drift, two-level-system episodes, recalibration renewals, physically coupled gate and readout errors, claim staleness, and data gaps, fitted to the DAQEC marginals by least squares. Table I reports every fitted target statistic, matched or not, and Fig. 3 shows the marginals and the dynamics.

The match is partial, and the misses matter for how Sim-Fleet may be used. The *static marginals* are reasonable: KS statistics are 0.22, 0.29 and 0.21 on T_1, T_2 and readout error, with two-qubit gate error matching poorly (KS 0.65). The *dynamics* are not. Lag-1 autocorrelation of T_1 has the wrong sign (real -0.156, sim $+0.202$): the real series oscillates on a recalibration-driven cadence that the simulator's monotone decay does not reproduce. Day-over-day persistence error, which sets forecasting difficulty, is 86% too high in simulation (0.064 real vs. 0.119 sim), and the recalibration renewal interval is off by a factor of 6.3 (1.31 h real vs. 8.32 h sim). These were all terms in the fit objective, so they are failures of the fit rather than unfitted extras. Consequently we treat Sim-Fleet as a stress environment with realistic marginals and *unrepresentative temporal structure*: it drives E1b and E2 model comparison (internal validity only), pretraining, E4b, E6a, and E7 gap injection, and no headline claim rests on it alone. Because the simulated series are both easier to persist against and more slowly recalibrated than the real ones, sim-only conclusions about forecasting difficulty and about staleness should not be transported to real hardware. A post-hoc refit with a recalibration-reset process (*Sim-Fleet-R*) reproduces these dynamics (lag-1 -0.20, persistence error 0.068, interval 1.40 h), and E7 is rerun on it. No live IBM access is used anywhere; the harvester is dry-run-tested and token-ready.

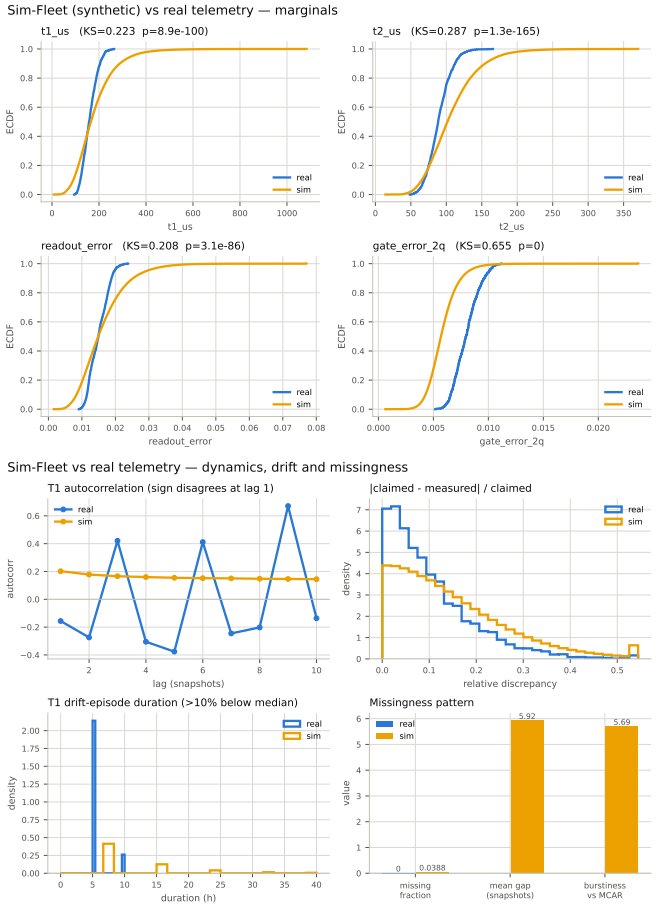


Fig. 3. Sim-Fleet vs. DAQEC. *Top:* marginal ECDFs. *Bottom:* T_1 autocorrelation and claim discrepancy. Statistics are in Table I.

Pre-registration and splits. Hypotheses, success bars, kill criteria, statistical protocol, and tuning budgets were frozen before any experiment run; deviations are declared inline and in the accompanying deviation register. Splits are committed with SHA-256 seals: E1a temporal 70/10/20 on DAQEC (horizons 1/3/7 days); E1b temporal on Sim-Fleet with four held-out devices (1/3/7/14 days); E3 three-fold device rotation; E4 device holdout; E5 platform holdout (Google never enters training); E6 replay windows; E8 temporal edge split. Normalization statistics are computed on training partitions only and stored inside the split files.

Baselines and fairness. Persistence, EWMA, per-series ARIMA, and LSTM, TGN-style [9], and causal-transformer forecasters share the same feature sets, split files, and evaluation pairs as LTN. Rather than fix capacity and tune the rest, we gave every learned model an *identical* tuning budget (50 Optuna [16] trials, TPE, validation NCRPS objective; persistence has no hyperparameters) with width/capacity itself a tuned dimension over a comparable range; the models are therefore *budget-matched*, not capacity-matched, and the tuned sizes span 17k to 282k trainable parameters (Table III, Params column). Consequently no capacity claim is made in LTN's favor: on the real split the strongest baseline (LSTM) is in fact several times *smaller* than LTN, while on the synthetic split the strongest baselines (TGN, transformer) are several times larger; LTN ties both without being capacity-privileged. Parameter matching

to $\pm 10\%$ *is* enforced only for the E2 ablation variants, which must be capacity-controlled to isolate a module's contribution. Our scale-free aggregate is NCRPS, defined as the Gaussian CRPS [12] divided by the training standard deviation of the observable; raw CRPS is reported per cell.

Statistics. Following the journal's convention we write *s-significant* where we intend the exact statistical sense, at $\alpha = 0.05$ ($p < 0.05$, or a 95% interval that excludes zero). Five seeds per stochastic model (deterministic baselines run once and are paired against each seed). The unit of analysis is the device×observable×horizon *cell*: we reduce each cell to its mean paired CRPS difference and run one Wilcoxon signed-rank test over cells, rather than testing individual forecast origins. Rolling origins within a series are strongly dependent (E1b averages 2,298 pairs per cell drawn from a single series), so treating them as independent inflates the effective sample size (it produces p-values as small as 10^{-133}). Holm correction controls multiplicity but not this dependence, which is why the unit, not the correction, is what we changed. Cliff's δ gives effect sizes [15]; Holm [14] is retained for the per-cell breakdowns. Confidence intervals are 10k-resample *cluster* bootstraps resampling (device, observable) series. We verified the choice of cluster variable: device-only clustering returns intervals *narrower* than iid for the deterministic baselines, the small-k degeneracy, so the series is the correct cluster.

V. Results

A. H1: forecasting device telemetry (E1, E2)

Table III reports NCRPS on the frozen test regions; Fig. 4 resolves horizons. On real DAQEC telemetry (E1a) LTN (0.292) **does not beat the baselines**: it is edged by LSTM (0.270) and TGN (0.279), and against the best baseline (LSTM) it is worse in 91.7% of the 36 device×observable×horizon cells (cell-level Wilcoxon over 36 cells, $p = 4.3 \times 10^{-7}$, Cliff's $\delta = 0.162$). Against the classical ARIMA (0.290) the honest verdict is a tie, not the loss the raw means suggest: the median Cliff's δ is 0.076 (negligible) and LTN wins 15 of the 36 cells. The 0.002 gap in the means is also smaller than the seed noise behind it, for reasons we take up next. On the large synthetic fleet (E1b: per-qubit, 12 training devices, 126 training days) the four neural models cluster within 0.9% (transformer/LSTM 0.310, TGN 0.311, LTN 0.312); LTN is nominally last among them but clears the classical floor, and the effect sizes are negligible even where the tests reject (cell-level Wilcoxon vs. the transformer over 240 cells, $p = 8.7 \times 10^{-24}$, Cliff's $\delta = 0.022$, LTN worse in 78.8% of cells).

Widening the intervals to respect the series structure changes three comparisons, one in LTN's favour and two against it. On E1b the paired LTN-versus-TGN difference (-0.0010) no longer excludes zero once clusters are resampled ($[-0.0026, +0.0005]$, against $[-0.0018, -0.0003]$ iid), so a finding against LTN becomes a tie. On E1a, however, LTN's advantages over the Gaussian-process ($+0.0455$) and Kalman ($+0.0488$) baselines of §V-C both cease to exclude zero ($[-0.0106, +0.1076]$ and $[-0.0042, +0.1072]$, against iid intervals that excluded it), so LTN loses two nominal wins. Every finding that LSTM and the transformer beat LTN survives. On E1a the clustered marginal intervals of all eight models overlap, including persistence, so that split does not separate any of the eight pre-registered arms from one another; it does separate all of them from the monotone-degradation baselines of §V-C. The pre-registered *beats the best baseline* bar for H1 is therefore **not met on either split**; on real data LTN is third of the four learned models. We conclude the physics-structured representation carries no point-forecasting advantage, and any justification must come from the state semantics evaluated under H2 and H3.

One seed, not one model. LTN's E1a mean is not a typical run. Its five seeds are 0.277, 0.279, 0.281, 0.288 and 0.338: four cluster inside 0.012 while one diverges, and that single seed alone moves the mean from 0.281 to 0.292 (robust $z = 10.2$). The effect is not unique to LTN, because the transformer diverges on one E1a seed too (0.333 against a median of 0.294), but it is largest for LTN, whose seed standard deviation is ± 0.026. We keep the pre-registered mean as the headline everywhere (over ten seeds, five added post hoc, it is 0.286, still third), and flag each conclusion whose *sign* depends on that seed; the E1b split shows no such instability (± 0.003).

The ablations (Table IV, Fig. 5) are exactly such a case. Taken at face value every E1a ablation is nominally *better* than the full model, by 0.011 (neural ODE [10] and −graph) down to 0.006 (−physics-γ). Every one of those advantages is manufactured by the diverging seed: comparing seed *medians* instead reverses all four signs, and the full model then leads by 0.001 (neural ODE), 0.000 (−graph), 0.004 (−Gaussian) and 0.001 (−physics-γ), negligible in the other direction. The claim that survives either way is the null one: no ablation *reproducibly* changes accuracy, and none reproducibly helps. In particular the exact Lindblad exponential and a parameter-matched neural ODE are indistinguishable (0.281 vs. 0.282 by median), so the Lindblad form buys nothing over a generic learnable ODE for forecasting, a conclusion the robust statistics support more cleanly than the means did. A dimension sweep of v1 on E1b is equally flat: $d \in \{2, 4, 8\}$ give 0.309/0.309/0.309, so the Hilbert-space dimension does not drive accuracy either. A v2 sweep on E1a (§V-G) agrees, but the two differ in model, split and dimensions, so they are consistent rather than strictly independent replications.

Where the variance comes from, and a mechanism we had to withdraw. The divergence is attributable to the learned Hamiltonian, but the mechanism we first proposed for *why* is contradicted by measurement, and we report the retraction rather than the hypothesis.

The hypothesis was this. Between observations LTN integrates $\exp(\mathcal{L}_v \Delta t)$ over gaps reaching tens of grid units, and the learned Hamiltonian enters $\mathcal{L}_v$ through the anti-Hermitian commutator $-i[H_v, \cdot]$, a purely oscillatory term that contributes no contraction. Nothing then damps the parameter-sensitivities of the exponential as the gap grows, and the Hamiltonian is not small once trained: on the E1a drift slice $\|H_v\|_F$ has median 6.9, inflating the generator norm from 0.70 to 19.6 (21.8× per generator). Large argument with no decay is the classic regime for a non-normal generator to amplify sensitivities.

To test it we captured every Lindblad generator each fitted E1a model actually used (drift steps and evidence-free horizon rolls, 15,120 Liouvillians in total) and measured seventeen

TABLE II
DATA INVENTORY. SIM ROWS ARE SYNTHETIC.

Source	Device	Events	Ent.	Obs.	Span	Type
sim (16 devices)[b]	16 devices	1,929,938	23–247	5	2026-01-01–2026-06-29	SIM
daqec[a]	ibm_brisbane	4,788	4	8	2025-01-15–2025-01-28	real
daqec[a]	ibm_kyoto	4,788	4	8	2025-01-15–2025-01-28	real
daqec[a]	ibm_osaka	4,788	4	8	2025-01-15–2025-01-28	real
google_qec	google_105q	11,604	119	3	2024-08-01–2024-08-18	real
google_qec	google_color	6,720	29	3	2024-12-01–2024-12-20	real
google_qec	google_dynamic	5,369	86	3	2024-12-01–2024-12-11	real
google_qec	google_sycamore_2022	2,079	30	3	2022-07-01–2022-07-06	real
silicon	simos_a	1,257	3	4	2022-06-06–2023-02-18	real
silicon	simos_b	2,166	3	4	2022-06-02–2023-05-31	real
silicon	simos_c	2,580	3	4	2022-09-01–2023-07-24	real

[a] DAQEC telemetry columns are used; the dataset's own experimental conclusions are not relied upon. [b] The 16 synthetic devices are summarized on one row; the per-device inventory is in the accompanying artifact. Forecast origins (series × origin time; train/val/test): E1a and E2, DAQEC 1,248/576/1,152; E1b and E2, Sim-Fleet 492,448/119,916/192,612; E3, 168 labeled DAQEC snapshots per held-out device; E4a, `ibm_osaka` adapted on 144–480 and evaluated on 384; E4b, four held-out Sim-Fleet devices adapted on 5,916–61,171 and evaluated on 69,217; E5, silicon 4,168 for training and DAQEC 1,968 for early stopping. Google contributes none: E5 ranks its 55 subgraphs zero-shot.

conditioning diagnostics per seed: Henrici departure from normality, eigenvector condition number $\kappa_2(V)$, numerical abscissa, spectral gap, commutator norms, and gradient norms at the fitted parameters. **The mechanism fails.** On 16 of the 17 diagnostics the diverging seed ranks fourth or fifth of five, and on all 15 that measure non-normality it ranks fourth or fifth. It has the *smallest* Henrici departure (0.227 against a median 2.07 for the other four), the smallest $\|H_v\|_F$ (3.76 vs. 8.32), the smallest numerical abscissa (0.022 vs. 0.385), and the smallest gradient norms (mean 1.05 vs. 1.23–1.88; maximum 1.57 vs. 2.19–4.29). The single diagnostic on which it ranks second, the ratio of Hamiltonian to dissipative content, is topped by seed 4, the *best*-performing run. With five seeds no correlation could be s-significant ($|\rho_s| \le 0.6$, $p \ge 0.28$ throughout), which is why the rank of the diverging seed is the readable signal; it points the wrong way.

A second check dissolves the framing as well: the $-$Hamiltonian ablation does not make the generator normal. v1's jump dictionary is itself non-normal, so zeroing H_v leaves a *more* non-normal dissipator on all five seeds (relative departure 0.013–0.161 rises to 0.205–0.490). Only the repaired model of §V-G is actually normal ($\kappa_2(V) = 1.000000$ exactly). The ablation therefore isolates H_v *present versus absent*, not *non-normal versus normal*.

We state the limit of this evidence. These are properties of the fitted parameters; no per-epoch checkpoints were retained, so a transient instability during training cannot be inspected retrospectively. That cuts both ways, because the original mechanism was also a claim about a trajectory we never measured, but every quantity we can measure points away from it. The causal attribution below stands on its own and does not depend on the mechanism.

The $-$Hamiltonian ablation (Table IV) holds the tuned configuration, the parameter count and every other module fixed, and only zeroes H_v in the generator; because it reuses the same seeds, it supports a *paired* comparison that a variance ratio cannot. The result is unambiguous. Seed 3, the diverging run, moves $0.338 \to 0.274$ when H_v is switched off, while the four healthy seeds move by at most 0.007 and the median is untouched ($0.2808 \to 0.2809$). The Hamiltonian is what diverges; it is not what forecasts.

Two cautions keep this honest. First, we make no claim of an

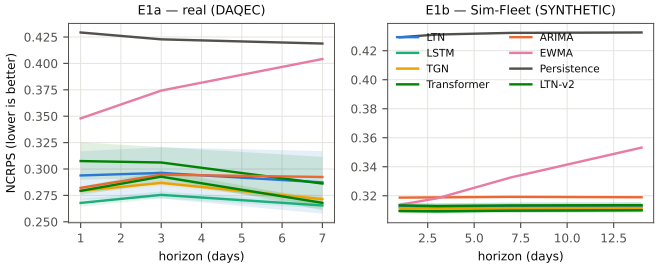


Fig. 4. NCRPS vs. horizon on real (E1a) and synthetic (E1b) telemetry.

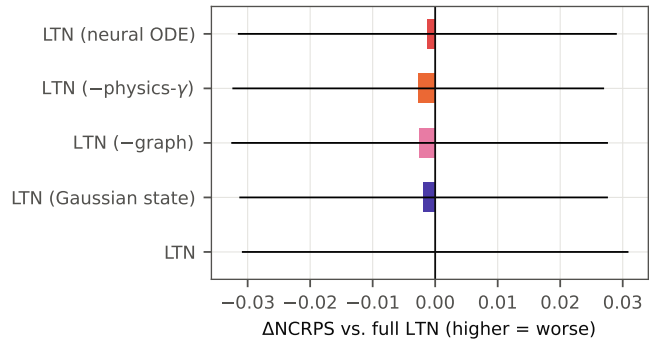


Fig. 5. Ablation deltas vs. full LTN (E1b, bootstrap CIs).

s-significant variance *ratio*: the seed standard deviation drops $\pm 0.026 \to \pm 0.005$, and to the repaired model's ± 0.002, but no variance test rejects, at five seeds (Levene $p = 0.38$ and 0.27, bootstrap ratio CI $[0.9, 59]$) or at ten ($p = 0.44$ and 0.37), because none of the new runs diverges. The evidence here is the paired seed and the disappearance of the divergence ($1/10$ runs to $0/10$), not a p-value on a spread. Second, $-$Hamiltonian does not reach the repaired model's stability (± 0.005 vs. ± 0.002), so H_v is the cause of the divergence but not the whole account of the spread; the bounded clock and the unital dictionary of §V-G carry the remainder. The rest of the ablation set agrees: the two variants that drop H_v as a side effect are the steadiest of the five (± 0.002 neural ODE, ± 0.003 →Gaussian) and the two least stable retain it (± 0.026 full, ± 0.016 −physics-γ), with $-$graph the exception at ± 0.005. We therefore make no forecasting claim for the density-matrix structure; if the Lindblad form has value it is its interpretable, CPTP-guaranteed semantics, not accuracy.

TABLE III
MAIN FORECASTING RESULTS (NCRPS, LOWER IS BETTER; EQUAL TUNING BUDGETS). INTERVALS ARE 95% *CLUSTER* BOOTSTRAP CIS RESAMPLING (DEVICE, OBSERVABLE) SERIES RATHER THAN INDIVIDUAL FORECAST ORIGINS, BECAUSE ROLLING ORIGINS WITHIN A SERIES ARE NOT INDEPENDENT; THEY ARE 1.7–4.6× WIDER THAN THE IID INTERVALS. PRE-REGISTERED ARMS ARE LISTED FIRST; LTN-V2 IS THE POST-HOC REPAIRED MODEL OF §V-G, REPORTED UNDER THE SAME PROTOCOL BUT SEPARATED BECAUSE IT WAS NOT PRE-REGISTERED. IT IS NOT A SEPARATE BASELINE, AND ON E1A ITS MEAN GAIN COMES FROM REMOVING A DIVERGING SEED RATHER THAN FROM BETTER FORECASTS.

Model	NCRPS [95% CI]	Seed med. (IQR)	Seeds	Params
E1a (real, DAQEC)				
LSTM	**0.270 [0.223, 0.319]**	0.269 (0.006)	5	16,808
TGN	0.279 [0.231, 0.334]	0.278 (0.004)	5	208,190
ARIMA	0.290 [0.231, 0.347]	–	1	–
LTN	0.292 [0.248, 0.344]	0.281 (0.010)	5	77,860
Transformer	0.300 [0.245, 0.357]	0.294 (0.004)	5	26,144
EWMA	0.375 [0.287, 0.463]	–	1	–
Persistence	0.424 [0.312, 0.548]	–	1	–
post-hoc, not pre-registered				
LTN-v2 (§V-G)	0.280 [0.229, 0.336]	0.281 (0.002)	5	67,852
E1b (Sim-Fleet, synthetic)				
Transformer	**0.310 [0.280, 0.341]**	0.309 (0.001)	5	259,622
LSTM	0.310 [0.281, 0.340]	0.310 (<0.001)	5	36,186
TGN	0.311 [0.282, 0.343]	0.311 (0.001)	5	282,196
LTN	0.312 [0.283, 0.344]	0.314 (0.004)	5	79,616
ARIMA	0.319 [0.289, 0.353]	–	1	–
EWMA	0.329 [0.298, 0.362]	–	1	–
Persistence	0.431 [0.390, 0.474]	–	1	–
post-hoc, not pre-registered				
LTN-v2 (§V-G)	0.313 [0.285, 0.344]	0.313 (0.002)	5	89,138

B. Computational cost

The measurements in this subsection are *post-hoc*; they were not pre-registered. The representation is expensive, and because it does not buy accuracy the cost belongs in the record. Training times are recovered from the campaign's own run records rather than re-measured; inference figures are medians over 100 repeats after warmup on one NVIDIA L40S.

Training. On real telemetry (E1a) LTN takes a median 138 s against LSTM's 11.3 s, a factor of 12; LTN-v2 takes 182 s. On the synthetic fleet (E1b) LTN takes 878 s against LSTM's 225 s, a factor of 3.9, and LTN-v2 takes 4,538 s, a factor of 20.

Inference. On the E1b workload (121 entities, 384 edges) one forecasting step costs LTN 5.64 ms on GPU against LSTM's 0.14 ms and TGN's 0.37 ms, and peak memory is 412 MiB against 9.6 MiB and 13.4 MiB. LTN is therefore roughly 40× slower and 43× larger in memory than the baseline that outperforms it.

Scaling in d. The propagator builds a $d^2 \times d^2$ superoperator and exponentiates it, so cost grows as $O(d^6)$ per node in time and $O(d^4)$ in memory. We did not characterize the empirical exponent, and we do not claim one. The analytic rate alone is enough for the design conclusion, because it would be a serious constraint only if accuracy improved with d: §V-G finds NCRPS flat across $d \in \{4, 5, 8\}$, and §V-I finds the forecast largely bypasses the state. The density-matrix propagator is thus the most expensive component of the model, scales worst in the one parameter that governs the trust state's expressiveness, and is the component the evidence says contributes least.

C. Comparison against classical reliability models

Persistence, EWMA and ARIMA are standard forecasting controls but they do not represent the reliability and degradation literature. We therefore added five classical reliability models, each fitted per series and each given the same 50-trial Optuna budget as every other baseline, running through the identical evaluation path on the same frozen splits (Table V). These models are *post-hoc*; they were not pre-registered.

On real telemetry the two state-space and kernel methods are competitive: a Gaussian-process degradation model reaches NCRPS 0.338 and a local-trend Kalman filter 0.341, against LTN's 0.292 and LSTM's 0.270, placing them between the transformer (0.300) and EWMA (0.375). We do not read that mean ordering as a win. Under the cell-level test we apply everywhere else, LTN separates from neither ($p = 0.19$ against the Gaussian process, $p = 0.13$ against the Kalman filter), and LTN is the better model in only 17 of 36 cells in each case: the mean gap is carried by a minority of high-error cells, not by a consistent advantage. The three remaining families do considerably worse (switching state-space 0.512, Wiener 0.541, gamma-process 0.582), and LTN does separate from all three under the same test ($p < 10^{-9}$ in each case, better in 94–97% of cells). For the two degradation models the reason is structural rather than a tuning failure: Wiener and gamma processes assume a monotone degradation path, whereas recalibration repeatedly resets this telemetry, so the monotonicity assumption is violated by construction. We have no equally specific explanation for the switching state-space result and do not offer one. We report them because a reliability audience will ask, and because a model failing for a stated structural reason is more informative than its omission.

The defensible summary is narrower than a win. LTN separates cleanly from the three structurally misspecified families, is statistically indistinguishable from the two well-specified ones, and is itself beaten by the generic recurrent baselines. Nothing here implicates the density matrix either way, because §V-I shows the forecast largely bypasses the state.

We report this comparison on the real split only. These models are fitted independently per series, so the cost of an equal 50-trial tuning budget grows with the number of series, and the synthetic fleet has two orders of magnitude more of them than DAQEC. We consider the real split the more informative one for a reliability audience in any case, and we prefer a complete comparison on real telemetry to a budget-starved one on both.

Change-point detection (E3; post-hoc). Drift detectors are not forecasters, because they emit an anomaly score rather than a predictive distribution, so they belong on E3's divergence-anticipation task rather than in the NCRPS table. Under the identical fold rotation, labels and threshold, a causal CUSUM reaches 0.461 mean AUROC and a causal Page–Hinkley detector 0.579; an *offline* binary segmentation, which sees the whole series and is therefore an optimistic bound rather than a peer, reaches 0.547. All are far below the pre-registered 0.75 bar, and Page–Hinkley merely matches the trivial last-discrepancy carry-forward (0.58). Methods purpose-built for detecting change also fail here. That strengthens rather than weakens the conclusion in §V-E: at this recalibration cadence the divergence signal is absent for every method we tried, not merely absent for ours.

TABLE IV

MODULE ABLATIONS (E2), NCRPS (SEED MEANS, AS PRE-REGISTERED, EXCEPT THE −HAMILTONIAN ROW, WHICH IS A POST-HOC CONTROLLED ABLATION ADDED AFTER THE VARIANCE FAILURE WAS OBSERVED; σ_{seed} IS THE SPREAD OVER 5 SEEDS). EVERY E1A ABLATION IS NOMINALLY BETTER THAN THE FULL MODEL, BUT EACH ADVANTAGE IS AN ARTIFACT OF ONE DIVERGING LTN SEED: ON SEED MEDIANS (Δ_{med}) ALL FOUR SIGNS REVERSE AND THE FULL MODEL LEADS BY ≤ 0.004. NO ABLATION REPRODUCIBLY CHANGES ACCURACY IN EITHER DIRECTION. −HAMILTONIAN IS THE CONTROLLED TEST OF THE VARIANCE DIAGNOSIS (§V-A): ZEROING H_v ALONE LEAVES THE MEDIAN UNTOUCHED (0.281) WHILE THE DIVERGING SEED RECOVERS, $0.338 \rightarrow 0.274$.

	E1a (real)						E1b (sim)			
Variant	NCRPS	Med.	IQR	Δ	Δ_{med}	σ_{seed}	NCRPS	Med.	IQR	Δ
LTN (full)	0.292	0.281	0.010	–	–	0.026	0.313	0.313	0.003	–
−Lindblad → neural ODE	0.281	0.282	0.003	−0.011	+0.001	0.002	0.311	0.312	0.004	−0.001
−physics-γ	0.287	0.282	0.005	−0.006	+0.001	0.016	0.310	0.309	0.002	−0.003
−DM → Gaussian state	0.285	0.285	0.001	−0.007	+0.004	0.003	0.311	0.310	0.001	−0.002
−graph	0.282	0.281	0.006	−0.011	+0.000	0.005	0.310	0.310	0.002	−0.002
−Hamiltonian	0.281	0.281	0.006	−0.012	+0.000	0.005	–	–	–	–

TABLE V

CLASSICAL RELIABILITY AND DEGRADATION BASELINES ON REAL TELEMETRY (E1A), ADDED POST-HOC. EACH IS FITTED PER SERIES WITH THE SAME 50-TRIAL TUNING BUDGET AS EVERY OTHER BASELINE. PRE-REGISTERED ARMS ARE LISTED FIRST, FOR REFERENCE. WIENER AND GAMMA MODELS ASSUME MONOTONE DEGRADATION, WHICH RECALIBRATION VIOLATES.

Model	Family	NCRPS
pre-registered (repeated from Table III)		
LSTM	recurrent	**0.270**
TGN	temporal graph	0.279
ARIMA	classical statistical	0.290
LTN	density-matrix state	0.292
Transformer	attention	0.300
EWMA	smoothing	0.375
Persistence	trivial	0.424
post-hoc, not pre-registered		
Gaussian process	GP degradation	0.338
Kalman (local trend)	linear-Gaussian state space	0.341
Switching state space	hidden Markov	0.512
Wiener process	degradation (monotone)	0.541
Gamma process	degradation (monotone)	0.582

D. H2: transfer (E4, E5)

Cold start on a held-out real device (E4a, `ibm_osaka`; Fig. 6 left): with sim-pretraining and fine-tuning on the two peer devices, LTN adapted on **one day** of target-device data reaches NCRPS 0.360 (95% CI $[0.304, 0.417]$), below the 0.545 ($[0.469, 0.628]$) that a from-scratch model reaches at **seven days**. The paired contrast is 0.186 NCRPS (95% CI $[0.063, 0.319]$, cluster bootstrap over the 12 observable×horizon series; positive on all five seeds). This is a $\geq 7\times$ data-efficiency gain against the pre-registered $\geq 2.3\times$ bar, and even the zero-shot pretrained model (0.515) beats scratch at any tested k. We state the interval rather than the point estimate because its lower end is a third of the point contrast: the direction is secure, the magnitude is not. On the synthetic fleet (E4b, four held-out devices) the effect saturates: pretrained zero-shot (0.309) already beats scratch with 30 days of target data (0.526), clearing the pre-registered $\geq 4\times$ bar. Because sim-to-sim transfer shares the generative process, however, E4a is the binding evidence. As pre-registered this is the paper's one positive result; the post-hoc baseline comparison below reassigns it from the architecture to pretraining.

Scope of E4. Both arms are LTN: as pre-registered, E4 varies *initialization*, not architecture. It therefore establishes that pretraining transfers for this model, not that the density-matrix state is what transfers. To find out which, we ran the identical pretrain-and-fine-tune protocol for the sequence baselines. The result removes the claim, and we report it in §V-D.

Do the baselines transfer as well? (post-hoc) E4 as pre-registered cannot separate *pretraining works* from *the density-matrix state transfers*. We therefore ran the identical protocol, with the same pretraining corpus, same masked-reconstruction objective, same tuning budget, same fine-tuning schedule and adaptation budgets, for LSTM, TGN, and a causal transformer (Table VI). Every family uses the five seeds of the E4a run reported above, so the LTN rows there are that run and the comparison is seed-matched. All four families run at the same E4 default hyper-parameters; we did not additionally sweep per-family tuned architectures, so the baselines are, if anything, handicapped relative to their own optima. This experiment is post-hoc; it was not pre-registered.

The claim does not survive. Pretraining helps every family, and it helps LTN *least*: at k=0 the LSTM improves $0.621 \rightarrow 0.434$, TGN $0.604 \rightarrow 0.380$ and the transformer $0.596 \rightarrow 0.384$, against LTN's $0.581 \rightarrow 0.515$. Pretrained LTN is last of the four families at every adaptation budget tested (k=0, 1, 3, and 7 days), with the mean paired difference favouring the baseline in all twelve (baseline, k) cells, though 0–1 of the 5 seeds favour LTN within each cell. With five seeds no individual comparison is s-significant ($p \geq 0.0625$, the minimum attainable at n=5, reached in five cells), so what we claim is that LTN is never ahead on average, not that it is s-significantly behind.

The $\geq 7\times$ figure is likewise an artifact of its reference arm. It divides the target data that *LTN-from-scratch* needs by what pretrained LTN needs, and LTN-from-scratch does not improve between one and seven days (0.546, 0.546 and 0.545). This suggests an under-optimized arm at these small adaptation budgets rather than a genuinely data-hungry model. A competitive reference changes the picture: the plain causal transformer, trained from scratch with no pretraining at all, reaches 0.346 on one day, which already beats pretrained LTN on one day (0.360). Recomputing the efficiency ratio against the best from-scratch arm anywhere in the table (that transformer at 0.339, reached with seven days), pretrained LTN never matches it (its best is 0.349), so against this reference it misses the pre-registered $\geq 2.3\times$ bar, while pretrained LSTM needs three days ($2.3\times$) and TGN and the transformer one ($7.0\times$). Every family clears the original bar zero-shot.

We therefore withdraw the transfer result as evidence for the density-matrix representation. What survives is narrower and belongs to the pretraining corpus, not to the architecture: cross-platform masked pretraining on synthetic and peer-device telemetry substantially improves cold-start forecasting for *every* temporal model we tested, and least for ours.

TABLE VI

COLD-START TRANSFER UNDER AN IDENTICAL PRETRAIN-AND-FINE-TUNE PROTOCOL (E4A, REAL, IBM_OSAKA HELD OUT; POST-HOC, NOT PRE-REGISTERED). MEAN NCRPS OVER 5 SEEDS, MEDIAN IN PARENTHESES (IQR AT MOST 0.066), LOWER IS BETTER; k IS DAYS OF TARGET-DEVICE DATA. PRETRAINED LTN IS LAST OF THE FOUR FAMILIES AT EVERY k.

Family	Arm	k=0	k=1	k=3	k=7
LTN	pretrained	0.515 (0.523)	0.360 (0.353)	0.349 (0.338)	0.351 (0.343)
	scratch	0.581 (0.572)	0.546 (0.534)	0.546 (0.534)	0.545 (0.534)
LSTM	pretrained	0.434 (0.452)	0.342 (0.334)	0.328 (0.324)	0.328 (0.320)
	scratch	0.621 (0.622)	0.574 (0.570)	0.574 (0.570)	0.573 (0.569)
TGN	pretrained	**0.380** (0.360)	0.326 (0.329)	0.313 (0.313)	0.312 (0.312)
	scratch	0.604 (0.614)	0.522 (0.520)	0.523 (0.522)	0.523 (0.521)
Transformer	pretrained	0.384 (0.367)	**0.313** (0.310)	**0.299** (0.297)	**0.297** (0.292)
	scratch	0.596 (0.586)	0.346 (0.349)	0.344 (0.343)	0.339 (0.338)

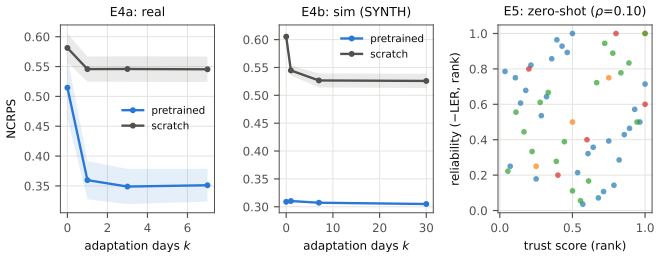


Fig. 6. Cold start (E4) and zero-shot platform ranking (E5).

Zero-shot platform ranking (E5): pre-registered demotion. Training on silicon, with IBM telemetry for early stopping, and ranking Google code placements by predicted error against decoded logical error per round yields pooled Spearman $\rho = 0.08$–0.14 across seeds (permutation $p \geq 0.29$, $n = 55$ subgraphs), below the pre-registered $\rho < 0.2$ demotion threshold. LTN's learned state is on par with the raw-DEM-intensity control ($\rho = 0.077$), and neither is s-significant. With the observable vocabularies this heterogeneous, cross-platform reliability transfer is *few-shot*, not zero-shot, and H2 rests on E4.

E. H3: trust semantics (E3, E6, E7)

Claim-divergence prediction (E3): pre-registered kill, reported. At DAQEC's near-continuous recalibration cadence (median per-run claim age 6.5 h, IQR 3.5–9.3 h, max 12.0 h; the freshest calibration available at any snapshot is a median 2.0 h old, IQR 1.2–2.9 h), relative claim-versus-probe divergence events are temporally indistinguishable from white noise, because no per-entity lag-1 autocorrelation leaves the white-noise band $\pm 1.96/\sqrt{42} = \pm 0.30$ (event indicator $\max|r_1| = 0.28$, median 0.13, over the 9 non-degenerate entity series; continuous relative discrepancy $\max|r_1| = 0.30$, median 0.10, over 36 series, each with $n = 42$ snapshots), and *no* predictor exceeds chance on the held-out device: LTN dissonance reaches 0.497 AUROC, an LTN state probe 0.511, and a staleness control 0.459. The best signal anywhere is a trivial last-discrepancy carry-forward at 0.58, still far below the pre-registered 0.75 bar. We conclude the divergence-anticipation claim is **untestable in this regime**: when claims are refreshed faster than the drift timescale, divergence is measurement noise. The claim would need telemetry with longer claim-aging (the situation our gap studies emulate) and we drop it here per the pre-registered kill criterion.

Operational value (E6). A policy selects, at each replay step, a w-qubit subgraph S on which to run a job; a candidate's realized success probability under the (hidden) ground-truth error rates is

$$p(S) = \prod_{q \in S} (1 - e_q^{\mathrm{ro}})(1 - e_q^{1\mathrm{q}})^{G_1} \prod_{(u,v) \in E(S)} (1 - e_{uv}^{2\mathrm{q}})^{G_2}, \quad (4)$$

with G_1=10 single- and G_2=5 two-qubit gate layers, and $e^{\mathrm{ro}}, e^{1\mathrm{q}}, e^{2\mathrm{q}}$ the readout, one- and two-qubit error rates. Regret is $\max_{S'} p(S') - p(S_{\mathrm{chosen}})$; policies see only claimed values or LTN forecasts, never the truth. In the truth-driven synthetic replay (E6a, Fig. 7) every informed policy cuts regret several-fold vs. random, but *no* pairwise difference between the informed policies (greedy-claimed, LTN-mean, LTN-LCB) separates at 95% bootstrap CIs over rounds (cumulative regret at w=12: LTN-LCB 0.226 vs. greedy 0.257). The pre-registered separation bar is not met: LTN forecasts convert to selection decisions *on par with* the calibration-greedy operational standard, not beyond it. Informed policies use a single trained checkpoint, so E6a isolates policy quality, not seed variance. The logged-arm DAQEC replay (declared secondary) is degenerate, because logged arms differ by $< 0.1\%$ in delivered success, so no policy separates there either.

Uncertainty quality (E7). We assess calibration with post-hoc temperature scaling fitted on validation. Temperature scaling was introduced for classifier logits [13]; the regression analogue we use rescales the predictive scale, $\sigma \mapsto t\,\sigma$ with

$$t = \arg\min_{t>0} \sum_i \left[\log(t\sigma_i) + \frac{(y_i - \mu_i)^2}{2t^2\sigma_i^2} \right], \quad (5)$$

the Gaussian negative log-likelihood on the validation split, optimized over $\log t \in [-10, 10]$ by bounded scalar search. It leaves μ and hence all sharpness-free accuracy metrics unchanged. On the synthetic fleet the forecasts are well calibrated (temperature 1.00, 90% coverage 0.906; Sim-Fleet-R: 1.00, 0.902); on real DAQEC they are systematically *under-dispersed* (temperature 1.07, 90% coverage 0.823 raw, 0.847 after scaling), a residual under-coverage we report rather than hide and attribute to the 14-day non-stationarity. The gap event study, run on Sim-Fleet (Fig. 8, left), returns a clean negative: during injected telemetry gaps the empirical forecast error grows monotonically to $+11.4\%$ by the fifth hidden snapshot, while the state entropy $S(\rho_t)$ stays flat ($|\Delta| \leq 0.6\%$) *and* the predictive σ (a declared post-hoc diagnostic) is equally flat ($|\Delta| \leq 1\%$). On Sim-Fleet-R (post-hoc; §IV; gaps of 8–40 h) the negative repeats: error rises to $+6.6\%$ as entropy falls ($+3.1\% \to +1.2\%$) and σ stays within 0.8%. The pre-registered entropy-tracks-error claim **fails**, and more broadly the trained model is overconfident under evidence starvation it never saw in training. Two mechanisms are identifiable, and both are architectural rather than incidental. First, with a horizon-conditioned forecast head, uncertainty need not route through the Lindblad state, so gradient pressure on the physics-coupled rates vanishes, and the training distribution contains no multi-snapshot all-entity gaps to correct it. Second, and worse than flatness: nothing constrains the learned jump operators to remain normal, and training pushes them away from it, so the fitted channel is non-unital and *purifying*. Rolled forward on E1a, the trained model's mean state entropy *decreases* down the horizon ($1.23 \to 1.10$ for $h = 1 \ldots 21$, falling for the large majority of states): the state does not merely fail to decohere, it

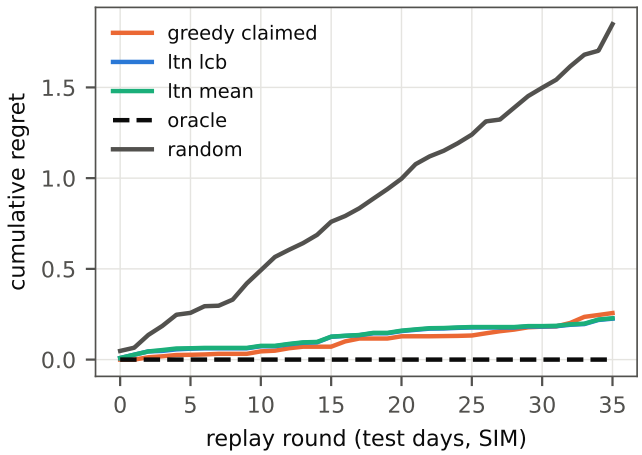


Fig. 7. Cumulative selection regret under replay (E6a, simulated fleet; policies see only claimed values or LTN forecasts, outcomes come from hidden truth).

recoheres (our term for growing purer without new evidence). The entropy channel could not have tracked growing error even had the head been willing to listen. Dissonance is likewise static around recalibration and divergence events (Fig. 8, right), consistent with E3. Forcing predictive σ to be read from the state alone is the architectural change these results motivate, and §V-G carries it out rather than leaving it to future work: it revives σ (which then rises through an evidence gap where v1's was flat; on Sim-Fleet-R only at gap onset, $+2.5\%$) but not the entropy signature, because the evidence channel, not the drift, is what governs the state through a gap.

F. Domain generality (E8)

To test whether the architecture is domain-bound, we drop DM-State and Lindblad decay into a DGTEN-style pipeline on Bitcoin-OTC and Bitcoin-Alpha (edge-weight prediction over 40 temporal windows) and compare against an equal-parameter vector-GRU control in the identical pipeline (Table VII). As in §II, E8 is an internal architecture ablation, not a re-benchmarking of DGTEN. At the pooled mean the density-matrix state improves both RMSE and sign-AUROC on both graphs. The two metrics do not, however, carry equal weight. The RMSE advantage holds on every seed pairing: the worst LTN seed mean beats the best control seed mean on both graphs. The sign-AUROC advantage does *not* survive at seed resolution, because the control's seed 4 beats two LTN seeds on Bitcoin-Alpha and one on Bitcoin-OTC, and that same seed is a large positive outlier for the control on both graphs, which is equally consistent with the control being under-trained on its other four seeds. We therefore read E8 as suggestive evidence on RMSE that the representation is not domain-bound, and we do not claim the AUROC result. This is the same single-seed pathology we police elsewhere in the paper, and it deserves the same treatment. Per-observable and per-device result tables and additional regret widths ($w \in \{5, 27\}$) are provided in the accompanying artifact.

G. Testing the diagnoses: a repaired model

Two of our negatives came with a mechanism attached: the seed variance was blamed on the learned Hamiltonian, a causal attribution that survives even though the non-normality

TABLE VII
DOMAIN GENERALITY (E8) ON BITCOIN TRUST GRAPHS: AN INTERNAL ARCHITECTURE ABLATION, NOT A RE-BENCHMARKING OF DGTEN.

Dataset	Model	RMSE	sign-AUROC
Alpha	LTN (DM+Lindblad)	0.157	0.581
Alpha	vector-GRU control	0.266	0.425
OTC	LTN (DM+Lindblad)	0.174	0.628
OTC	vector-GRU control	0.303	0.521

account of *why* does not (§V-A), and the failed entropy-growth signature on a readout that lets uncertainty reach the horizon without passing through the state (§V-E). A mechanism that cannot be falsified is not a scientific claim, so we built the model those diagnoses imply and re-ran the affected experiments under the same protocol as every other model: the same frozen splits, the same 50-trial Optuna budget over the same search space, and the same five seeds for the headline runs (the d sweep uses three per arm, as v1's did; the E1b tuning completed 49 of its 50 trials after a GPU failure, a shortfall that can only disadvantage v2). It changes four things, which is precisely why it can corroborate a diagnosis but not isolate one. *(i) Born-rule readout*: $\mu_c = a_c \operatorname{tr}(\rho_h O_c) + b_c$ and $\sigma_c = s_c \sqrt{\operatorname{tr}(\rho_h O_c^2) - \operatorname{tr}(\rho_h O_c)^2}$ for learned Hermitian O_c, so σ is a functional of the state alone, with no horizon or carry path; uncertainty can then only grow if the drift mixes the state, which forces $\partial\sigma/\partial\gamma \neq 0$. *(ii) Unital dissipative drift*: H_v is deleted and the jump dictionary is restricted to Hermitian (hence normal, hence unital) operators. Four properties must be distinguished here, because only two are theorems. *Unitality* is structural, holding for any parameters, and *entropy non-decrease* follows from it by majorization; both hold for any learned rates. *Uniqueness of the fixed point* does not: it requires the $\{L_k\}$ to generate the full matrix algebra, and Hermitian dephasing jumps are a standing counterexample, being unital and entropy-non-decreasing while fixing every diagonal state. For these fits uniqueness holds *empirically*: the generator's kernel has dimension exactly 1 in all 25 fitted v2 checkpoints, rate-independently (the commutant of $\{L_k\}$ is trivial in $25/25$), and in all 2,520 generators the model uses on real E1a timelines, with a rank-determination margin of fourteen orders of magnitude. *Convergence to I/d* then follows, but as an empirical property of these fits, not a guarantee of the construction. This repairs an *observed* pathology, not a hypothetical one. Nothing in v1 constrains its jump operators to stay normal, and training drives them away from it ($\|[L_k, L_k^\dagger]\|$ reaches 0.5–1.4 across the fitted E1a seeds), which makes the learned channel non-unital and therefore *purifying*. The consequence is a trust model running in reverse: rolling that model forward, mean state entropy *falls* along the horizon ($1.23 \to 1.10$ over $h = 1 \ldots 21$, decreasing for the large majority of states). v1 grows *more* certain the further it extrapolates from evidence, the exact opposite of the epistemic decoherence the model is named for, and an anti-signature to the one E7 sought. Restricting to Hermitian jumps makes the correct direction a structural property rather than something to hope training respects. *(iii) Saturating clock*: the gap enters as $\tau(\Delta t) = \tau_{\max}(1 - e^{-\Delta t/\tau_{\max}})$, bounding the matrix exponential's argument; because τ is concave this is the exact Lindblad solution in a warped clock, not in physical

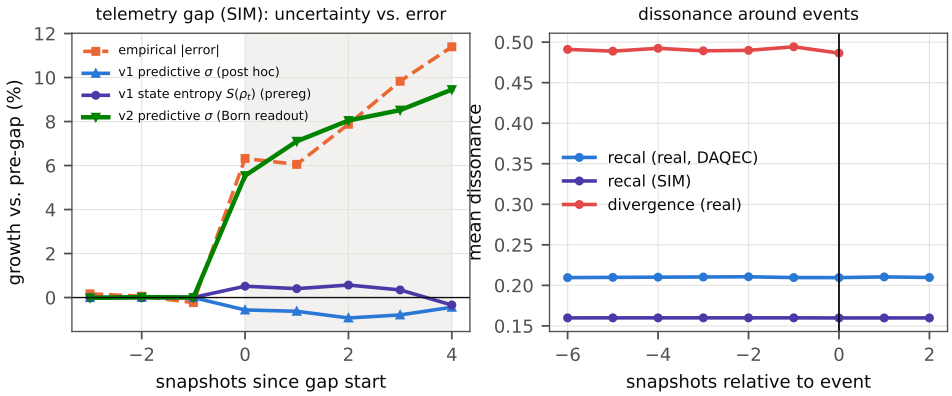


Fig. 8. Gap event study on Sim-Fleet. Left: during injected telemetry gaps (shaded) v1's empirical forecast error grows to $+11.4\%$ while its state entropy $S(\rho_t)$ (pre-registered) and predictive σ (post hoc) stay flat; the repaired model's σ (§V-G) is the one uncertainty curve that follows the error. Right: dissonance is static around recalibration and divergence events, consistent with the E3 null.

time. *(iv) Observable-aligned state*: $d = n_{\text{obs}}+1$.

The variance diagnosis is **confirmed**. On E1a the repaired model's seed standard deviation is ± 0.002 against ± 0.026, no seed diverges (worst run $0.338 \to 0.282$), and NCRPS improves to 0.280: level with TGN (cell-level Wilcoxon over 36 cells, $p = 0.88$, Cliff's $\delta = -0.04$), still tied with ARIMA but now on the better side of it, and much nearer LSTM than v1 was. We deliberately do not report this as a change in rank. None of it is new forecasting skill: against v1's four non-diverging seeds the difference is 0.280 vs. 0.281, which no test resolves (seed-level Welch $p = 0.66$; 0.19 over ten seeds). The entire movement is the deleted outlier, and a mean over five seeds is sensitive to one of them. E1b settles the point. That split never had a diverging seed to repair, and there the repair buys nothing: v2 and v1 are indistinguishable (median Cliff's $\delta = 0.002$; v2's mean is 0.001 higher, last among the learned models). A repair earns nothing where there is nothing to repair, which is exactly what we should expect and the reason we do not report it as an improvement. Attribution likewise rests on the controlled $-$Hamiltonian ablation of §V-A, not on this model, which moves four things at once and so can corroborate the diagnosis but never isolate it.

The readout diagnosis is **confirmed, and the pre-registered signature still fails**, a split verdict worth stating exactly. We measure it on the same injected gaps as v1 (Fig. 8), tracking each quantity as a percentage of its pre-gap level so the two models are read the same way. The predictive σ finally responds to evidence starvation: through a gap v2's σ rises at every hidden snapshot, from $+5.5\%$ at gap onset to $+9.4\%$ by the fifth, moving with the empirical error ($+9\%$ to $+11\%$), whereas v1's σ drifts slightly *down* over the same gaps (-0.6% to -0.4%) and its entropy stays flat. Uncertainty that could previously ignore the state now cannot, which is precisely what the Born-rule readout was built to force, and calibration improves with it (real: temperature $1.07 \to 1.01$, raw 90% coverage $0.823 \to 0.852$; simulated: $1.00 \to 0.99$, $0.906 \to 0.908$). Yet the pre-registered claim, that *entropy* tracks error, fails again, and for a reason the repair exposes rather than removes. v2's entropy does jump when a gap opens (evidence loss raises it $+12\%$ over the pre-gap level), but it then drifts back *down* as the gap deepens ($+12.2\%$ at onset to $+9.9\%$ at the fifth snapshot) while the error keeps climbing, so it does not track error. Decomposing one gap step of the trained model shows why: the unital drift raises entropy for every node (exactly the guarantee), but the evidence channel, which is CPTP yet not unital and is still applied through a gap because the snapshot is empty rather than absent, then removes more than the drift added. The guarantee we bought is real and it binds on the drift; the drift is simply not what governs the state through a gap. Making evidence unital under a null observation, so that seeing nothing cannot sharpen a belief, is the change this points to and the one we have not yet tested. We stop short of claiming σ *tracks* error: the across-position correlation rises from -0.24 to $+0.81$, but with six gap depths it is not resolvable from zero, and the empirical error is itself noisy. What is established is narrower and still worth having: the uncertainty channel is now connected to the state.

The representation diagnosis **does not survive**. If the density matrix were doing the forecasting once σ is forced through it, accuracy would have to respond to its dimension. It does not: on E1a, $d \in \{4, 5, 8\}$ (3 seeds each) give $0.279/0.279/0.280$, a spread of 0.0005 that is $0.22\times$ the within-d seed noise. We read this as equivalence rather than as a large p: the whole effect of doubling d is bounded well inside the noise of re-running the same d, which is the practical definition of not mattering. It echoes v1's flat sweep ($0.309/0.309/0.309$), though that one varied $d \in \{2, 4, 8\}$ on the synthetic split, so the two are consistent rather than strictly independent replications. The repaired model is therefore not a better forecaster, and we do not present it as one. Its value is negative evidence made sharper, and it defines what would have to change for a physics-structured state to earn its place: a task where the state, not the recurrent carry, is the thing being asked for.

H. Reliability endpoints

These endpoints are *post-hoc*; they were not pre-registered, and they re-score the pre-registered forecasts rather than adding new model runs. NCRPS measures forecast quality, not reliability. Because the forecasts are Gaussian, three reliability quantities follow in closed form from the stored (μ, σ) without retraining or refitting, so they cannot flatter any model relative to the numbers already reported (Table VIII).

SLA-violation probability. Writing q for a per-device, per-observable specification limit, $P(Q_{t+h}$ out of spec$) = \Phi((\mu -$

TABLE VIII

RELIABILITY ENDPOINTS DERIVED IN CLOSED FORM FROM THE STORED GAUSSIAN FORECASTS. (A) SLA-VIOLATION PROBABILITY $P(Q_{t+h}$ OUT OF SPEC) ON THE SYNTHETIC FLEET (E1B), SLA SET PER DEVICE–OBSERVABLE AT A 10% EMPIRICAL VIOLATION BASE RATE; MEAN ± S.D. OVER SEEDS. BRIER SKILL IS AGAINST THE CLIMATOLOGICAL BASE RATE (HIGHER IS BETTER; 0 = NO SKILL). $\Delta R(14)$ IS THE CHAINED-SURVIVAL ERROR AT THE LONGEST HORIZON, $R_{\text{pred}} - R_{\text{emp}}$ (NEGATIVE = PESSIMISTIC).

Model	Brier ↓	ECE ↓	Brier skill ↑	$\Delta R(14)$
Transformer	0.0626 ±0.0003	0.0080 ±0.0018	0.304	−0.001
LSTM	0.0626 ±0.0002	0.0055 ±0.0021	0.304	−0.006
ARIMA	0.0629	0.0119	0.300	+0.028
TGN	0.0632 ±0.0005	0.0072 ±0.0016	0.298	+0.010
LTN	0.0632 ±0.0007	0.0079 ±0.0046	0.297	−0.006
LTN-v2	0.0635 ±0.0006	0.0063 ±0.0008	0.294	−0.006
EWMA	0.0672	0.0419	0.252	−0.113
Persistence	0.0906	0.0728	−0.008	−0.128

$q)/\sigma)$ for error rates and $\Phi((q-\mu)/\sigma)$ for coherence times. We set q at the empirical quantile of the observed outcomes that puts 10% of mass out of spec, computed from y alone so it is identical for every model; LTN remains mid-pack and behind both the transformer and LSTM at 5%, 10% and 20%, though the ordering among the other arms shifts (ARIMA in particular falls from first to fifth as the threshold loosens). On the synthetic fleet LTN is mid-pack: a paired device-clustered bootstrap puts it s-significantly *worse* than the transformer ($+0.00063$, $[+0.00035, +0.00114]$) and LSTM ($+0.00062$, $[+0.00007, +0.00171]$), indistinguishable from TGN, ARIMA and LTN-v2, and better only than EWMA and persistence. The reliability endpoint therefore reproduces the H1 verdict rather than overturning it, which is the outcome we would expect if the two measure the same underlying forecast quality.

Reliability function and hazard. Chaining $(1 - p_{\text{viol}})$ across checkpoints gives $R(h)$; LTN's chained survival is mildly pessimistic at the longest horizon ($\Delta R(14) = -0.006$, against -0.001 for the transformer and -0.113, -0.128 for EWMA and persistence). We caution against a physical reading of the declining per-checkpoint hazard ($0.102 \to 0.048$ from h=1 to h=14): the checkpoint spacing widens over the same range, and the surviving cohort is disproportionately the stable device–observable pairs, so this is frailty, not a decreasing failure rate.

Expected wasted shots and availability. On the synthetic replay, with 4096 shots per job, informed selection recovers roughly 220 shots per job against random at width 12, a large and robust effect. Informed policies do not separate from one another: LTN-LCB leads greedy-claimed at widths 5 and 12 but trails it at width 27, and all three lie within ~ 14 shots of one another, out of 4096. This matches the E6 regret null and we claim no separation. On the real logged replay the lane is degenerate: random selection wastes 0.51 shots out of 4096 more than the oracle (0.0125%), and the informed policies waste marginally *more* (0.65 shots). There is no operational headroom in which any policy could demonstrate value on this data, which we report rather than omit.

I. Is the density matrix load-bearing? A state-sufficiency test

The flat d sweep admits a blunter explanation than "the state is redundant with the carry": that predictions never route through the state at all. The architecture maintains both a density matrix ρ and a GRU carry, and nothing in the design forces the readout to use the former. We test this directly. The experiment is *post-hoc and exploratory*; it was not pre-registered.

TABLE IX

DISCRETE-TIME RELIABILITY $R(h) = P$(NO SLA VIOLATION THROUGH h) AND HAZARD ON THE E1B HORIZON GRID, FIXED COHORT OF ORIGINS OBSERVED AT EVERY HORIZON. R_{emp} IS A PROPERTY OF THE DATA AND THE SLA, IDENTICAL FOR ALL MODELS; R_{pred} IS THE LTN FORECAST CHAINED UNDER CONDITIONAL INDEPENDENCE.

h	R_{pred}	R_{emp}	haz. pred	haz. emp
1	0.898	0.901	0.102	0.099
3	0.835	0.840	0.070	0.068
7	0.788	0.794	0.056	0.055
14	0.750	0.757	0.048	0.047

TABLE X

EXPECTED WASTED SHOTS PER JOB OF $N = 4096$ SHOTS, $N(1 - p(S))$, ON THE E6A SYNTHETIC REPLAY (w=12). INFORMED POLICIES ARE *SINGLE-SEED*; RANDOM IS AVERAGED OVER 5 SEEDS. THE EXCESS COLUMN IS THE SHOTS AN OPERATOR LOSES RELATIVE TO THE ORACLE SELECTION.

Policy	Availability	Wasted shots	Excess vs. oracle
oracle	0.6350	1495.1	0.0
ltn_lcb	0.6287	1520.7	25.7
ltn_mean	0.6287	1521.0	25.9
greedy_claimed	0.6279	1524.2	29.2
random	0.5751	1740.6	245.5

We use v2, where the test is clean: v2 sets the Hamiltonian to zero, so the carry is read by exactly one module, the residual $\text{resid}(\text{carry})$ added to μ, while σ is a functional of ρ alone. In v1 the carry also drives the Hamiltonian, so a carry-free v1 would confound with the $-$Hamiltonian ablation. One caveat bounds every reading below: the GRU is *written* from (x, ρ) at every step, so the carry is itself a function of the state's history. Evidence that the carry dominates therefore shows that the readout reaches the state through the GRU's real-vector compression, not that ρ is inert.

Three interventions are applied at readout to the trained checkpoints, with no retraining and no weight updates (Table XI). Zeroing the carry residual costs $+110\%$ NCRPS on E1a and $+81\%$ on E1b, on all five seeds of both splits. Permuting ρ across entities costs $+4.3\%$ on E1b (5/5 seeds, consistent sign); permuting the carry instead costs $+94.5\%$. A magnitude decomposition of the readout agrees: the carry residual is $2.9\times$ (E1a) and $3.0\times$ (E1b) larger than the Born term $a_c \,\text{tr}(\rho_h O_c)$ in mean absolute value, and supplies $5.0\times$ and $7.9\times$ more *between-entity* variation. The E1a permutation arms are under-powered and we do not quote them as evidence: with four entities per device the readout is nearly entity-constant, so a permutation has little to scramble, and both E1a permutation deltas sit inside seed noise without a consistent sign.

Two conclusions follow. First, the zero-initialization of the residual, which we had treated as a constraint, is not one: training moves the residual to roughly three times the magnitude of the Born term, and the carry path on μ is wide open. Second, on E1b the state is measurably load-bearing but does a small share of the work, roughly one part in twenty-two by the permutation contrast. It is not dead; it is also not what discriminates between entities. This is the most direct evidence in the paper that the density-matrix representation is not carrying the forecasting task, and it explains the null d sweep without appealing to redundancy. Retrained without the carry (post-hoc), so forecasting from ρ alone, the model still

TABLE XI
STATE-SUFFICIENCY TEST ON TRAINED LTN-V2 CHECKPOINTS (POST-HOC, NOT PRE-REGISTERED). READOUT-ONLY INTERVENTIONS, NO RETRAINING; MEAN NCRPS OVER FIVE SEEDS, LOWER IS BETTER. *SEEDS WORSE* COUNTS SEEDS ON WHICH THE INTERVENTION HURTS.

Split	Intervention	NCRPS	Δ	Seeds worse
E1a	none (baseline)	0.2800	–	–
	zero carry residual	0.5886	+110.2%	5 of 5
	permute ρ	0.2802	+0.07%	3 of 5
	permute carry	0.2794	−0.21%	2 of 5
E1b	none (baseline)	0.3133	–	–
	zero carry residual	0.5661	+80.7%	5 of 5
	permute ρ	0.3268	+4.31%	5 of 5
	permute carry	0.6093	+94.5%	5 of 5

trails LSTM on E1a (0.293, $p = 3 \times 10^{-4}$).

VI. LIMITATIONS AND ETHICS

Observational, not causal. LTN forecasts and ranks; it does not identify root causes of degradation. A trust score is a calibrated prediction about future telemetry and claim-versus-delivery consistency, never a diagnosis.

Failure modes without telemetry traces. Any failure mechanism that leaves no trace in the observed telemetry cannot lower a trust score directly. The architecture was designed so that such blindness would surface as *vacuity*, that is, state uncertainty growing under the learned Lindblad drift between observations, rather than as a confident score. E7 tests that design intent and refutes it for the trained model: entropy *falls* along the horizon ($1.23 \to 1.10$ for $h = 1 \ldots 21$, §V-G), so the original model answers evidence starvation with increasing confidence. The repaired model of §V-G restores entropy non-decrease structurally by making the drift unital, but we have not shown that the restored signal detects untraced failures; that remains an open claim, not a guarantee.

No live fleet access. This work uses no privileged or live IBM Quantum access. Real-hardware evidence comes from public archives: DAQEC (three IBM Eagle devices, 14 days, subgraph-granular), Google QEC releases (episodic, per-run), and a silicon spin-qubit dataset (three devices, episodic). Longitudinal depth beyond 14 days is provided by a *synthetic fleet simulator* whose generative processes are moment-matched to the DAQEC marginals; every table and figure labels simulated data explicitly, and no headline claim rests on simulation alone. The complete, dry-run-tested fleet harvester is provided with the code for groups with API access.

DAQEC provenance (a load-bearing caveat). All of our real IBM evidence (E1a, E3, E4a) rests on one archival dataset, DAQEC, which is a single-author, non-peer-reviewed deposit; its own headline conclusions include near-chance-level validation runs. We therefore use only its raw telemetry columns (claimed vs. probe-measured calibration, logical error rates, syndrome statistics), which are internally consistent and physically plausible, and we rely on none of its experimental conclusions. Independent replication on additional longitudinal IBM telemetry (which our token-ready harvester would collect) is the most important external-validity step for the real-data claims.

Real data granularity. DAQEC reports subgraph-averaged calibration, so real-data forecasting (E1a) is subgraph/device-granular; per-qubit claims are supported only on simulation (E1b) and on per-qubit noise intensities extracted from Google detector error models (E5 inputs).

Quantum-native formalism, classical execution. The state space and dynamics are quantum-mechanical (density matrices, CPTP channels, Lindblad generators) and the telemetry comes from real quantum hardware, but all computation is classical GPU linear algebra. We claim no quantum computational advantage anywhere.

Deviations from pre-registration. Deviations decided before the results were known (horizons shortened to the real data span, revised success bars, simulation-primary operational replay, and budget-matched rather than fixed-capacity baselines) are enumerated in the accompanying deviation register and flagged inline where they affect a claim. The register additionally carries a *post-freeze* block covering everything decided after the pre-registered results were known: the repaired model, the controlled $-$Hamiltonian ablation, and the further post-hoc analyses (state sufficiency, matched-pretraining transfer controls, classical reliability baselines, change-point detectors, reliability endpoints, and cost measurements). Roughly half the experiments reported here fall in that block. We consider the distinction load-bearing rather than procedural: the pre-registered arms test hypotheses fixed in advance, whereas the post-hoc analyses were designed once their targets were visible, and we make no claim that they carry equal evidential weight.

Ethics. All datasets are public under CC-BY-4.0 (or SNAP terms for E8); no human subjects are involved. Trust scores over shared quantum infrastructure could, if miscalibrated, unfairly disadvantage a vendor; our calibration analysis (E7) and vacuity-first treatment of missing telemetry are designed to make under-evidenced negative judgments visible as uncertainty rather than confident distrust.

VII. CONCLUSION

Trust in quantum hardware decays the way quantum information does: continuously, stochastically, and at rates the hardware itself tells us. LTN makes that observation operational through density-matrix trust states, learned physics-coupled Lindblad decay, CPTP evidence and propagation, and POVM readout, and it grounds them in public, multi-platform, real-hardware telemetry with pre-registered evaluation. The verdict is negative throughout, and that is the contribution's honest shape. On three IBM devices over 14 days, LTN does *not* match tuned recurrent baselines: it ranks third of the four learned models, ahead only of the transformer and level with the classical ARIMA, and no individual module is reproducibly load-bearing. The cold-start gain we initially read as the paper's one positive result does not belong to the architecture: under an identical pretrain-and-fine-tune protocol every sequence baseline transfers better, and LTN gains the least from pretraining of the four families. A direct state-sufficiency test locates the reason, namely that the forecast reaches the state mainly through the recurrent carry rather than through the density matrix. This also explains why accuracy is indifferent to the state's dimension. The trust-semantic payoffs of divergence anticipation and uncertainty-aware selection did not materialize in DAQEC, a benchmark whose sub-drift-timescale recalibration cadence makes the claim-staleness signal LTN models nearly absent. Where the

method should matter is precisely the opposite regime of sparsely calibrated devices, long claim-aging, and telemetry-silent failure modes, the regime that our gap-injection studies emulate and that live-fleet harvesting (through the token-ready harvester) would supply. We offer the schema, benchmark splits, harvester, and one-command reproduction as infrastructure for the community to extend toward that regime, and toward QKD trust topologies and trust-budgeted verification.

Appendix

Proof of Proposition 1

For diagonal ρ, all readout operations are simultaneously diagonalizable, giving (i) and (iv) by direct computation; (ii) is verified by substitution, $b_c + a_c u = (1-v)\mathrm{score}_c + v\,\mathrm{score}_c = \mathrm{score}_c$; (iii) is a parameter count. Two cautions follow from this statement and are not corollaries of it. First, "belief equals p" additionally requires the learned POVM to be the computational-basis projectors, which the construction does not enforce. Second, a column-stochastic map is a Markov transition, *not* a Bayesian evidence update: Bayes' rule $p_j \mapsto p_j \ell_j / \sum_k p_k \ell_k$ is projective rather than affine, so no p-independent matrix realizes it for a nontrivial likelihood.

Proof of Proposition 2

With $\rho = P_\rho \rho = \rho P_\rho$, the hypothesis $[K_i, L_j]P_\rho = 0$ gives $K_i L_j \rho = L_j K_i \rho$ and, by taking adjoints, $\rho L_j^\dagger K_i^\dagger = \rho K_i^\dagger L_j^\dagger$, so the two sandwiched sums agree. We do not claim the converse, and we explicitly do *not* claim that order-dependence is a coherence effect: two diagonal-preserving channels with non-commuting transition matrices already fuse order-dependently on diagonal states. At d=2, reset-to-$|1\rangle$ with Kraus $\{|1\rangle\langle 1|, |1\rangle\langle 2|\}$ and reset-to-$|2\rangle$ with $\{|2\rangle\langle 2|, |2\rangle\langle 1|\}$ are both trace-preserving, and the two composition orders give $|1\rangle\langle 1|$ and $|2\rangle\langle 2|$: maximally different, with identically zero off-diagonal mass.

Proof of Proposition 3

A unital trace-preserving map acts doubly stochastically on the spectrum, so $\mathrm{eig}(\Phi(\rho))$ is majorized by $\mathrm{eig}(\rho)$; $\mathrm{tr}\,\rho^2 = \sum_i \lambda_i^2$ is Schur-convex, hence $\mathrm{tr}\,\Phi(\rho)^2 \le \mathrm{tr}\,\rho^2$ and vacuity is non-decreasing. We note for contrast that the trace-distance route does *not* establish this: non-expansiveness [8] with $\sigma = I/d$ gives $\|\Phi(\rho) - I/d\|_1 \le \|\rho - I/d\|_1$, a statement about $\|\cdot\|_1$ and not about the purity functional v, and the two orderings genuinely disagree.[2] The general non-unital case depends on the channel's actual fixed point and is left open.

Conflict of Interest

The authors declare that they have no conflict of interest.

Data Availability

The universal telemetry schema, the evaluation pipeline, the token-ready harvester, the pre-registration, the deviation register, and the SHA-256-sealed splits are available at https://github.com/GreenPatterns/LTS (commit 4a87c7f), enabling one-command reproduction from public data. All primary datasets used are publicly available and are cited in the text.

[2] At d=3, $\rho_a = \mathrm{diag}(0.70, 0.15, 0.15)$ has $\|\rho_a - I/3\|_1 = 11/15$ and $\mathrm{tr}\,\rho_a^2 = 0.535$, while $\rho_b = \mathrm{diag}(0.65, 0.35, 0)$ is strictly closer to vacuous ($\|\rho_b - I/3\|_1 = 2/3$) yet strictly *more* pure ($\mathrm{tr}\,\rho_b^2 = 0.545$).

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